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SUBJECT CODE NO: - RNP-8001

FACULTY OF SCIENCE AND TECHNOLOGY

F. E. (NEP) [Group A & B] Mech, Civil, Comp. AI & ML, EEE

Examination (January/February-2025)

Applied Mathematics - I

[Time: 3:00 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve the following questions: (Any 10)

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- a) Which of the following functions can be expressed as a Maclaurin's series?
 - a) Any differentiable function.
 - b) Functions that are differentiable at $x=0$.
 - c) Only polynomial functions.
 - d) Any continuous
- b) Taylor's theorem is valid for which types of functions?
 - a) Continuous functions
 - b) Functions that is infinitely differentiable.
 - c) Only polynomial functions.
 - d) Functions with a finite number of derivatives question/multiple
- c) When evaluating $\lim_{x \rightarrow \infty} \frac{x}{e^x}$ which indeterminate form arises?
 - a) $0/0$
 - b) ∞ / ∞
 - c) $0 \times \infty$
 - d) 1^0
- d) By the Cauchy's n^{th} root test the series $\sum_{n=1}^{\infty} u_n$ is convergent if
 - a) $\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} < 1$
 - b) $\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} > 1$
 - c) $\lim_{n \rightarrow \infty} (u_n)^{\frac{1}{n}} = 1$
 - d) 0
- e) The maximum value of the function $\sin x \cdot \sin y \cdot \sin(x+y)$ is
 - a) $\frac{3}{2}$
 - b) $\frac{\sqrt{3}}{2}$
 - c) $\frac{3\sqrt{3}}{8}$
 - d) 1

- f) The Integrating Factor of $\frac{dy}{dx} + xy = x^3$ D.E is
 - a) $x^2y = c$
 - b) $x = c$
 - c) $e^{\frac{x^2}{2}}$
 - d) $x^2y - xy = c$
- g) The necessary & sufficient condition for D.E. to be exact D.E. is
 - a) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
 - b) $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$
 - c) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$
 - d) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$
- h) A circuit containing resistance R and inductance L in series with voltage source E then the D.E for current is
 - a) $Li + R\frac{di}{dt} = E$
 - b) $L\frac{di}{dt} + Ri = E$
 - c) $Li - R\frac{di}{dt} = E$
 - d) $Li + R\frac{di}{dt} = 0$
- i) If $u = \frac{x^2y^2z^2}{x^2+y^2+z^2}$ then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$ is equal to
 - a) 0
 - b) 1
 - c) 4u
 - d) u
- j) If u, v & w are functions of three independent variables x, y & z then the Jacobean is given by
 - a) $J = \frac{\partial(x,y,z)}{\partial(u,v,w)}$
 - b) $J = \frac{\partial(x,v,z)}{\partial(u,y,w)}$
 - c) $J = \frac{\partial(u,v,w)}{\partial(x,y,z)}$
 - d) $J = \frac{\partial(z,y,x)}{\partial(w,y,x)}$
- k) A body of mass m falls from rest under gravity, in a fluid whose resistance to motion at any time t is mk times its velocity where k is constant, the D.E. of motion is
 - a) $\frac{dv}{dt} = -g - kv$
 - b) $\frac{dv}{dt} = g - kv$
 - c) $\frac{dv}{dt} = -g + kv$
 - d) $\frac{dv}{dt} = mg - mkv$
- l) If $u = \log(x^3 + y^3)$ then $\frac{\partial u}{\partial x}$ is
 - a) $\frac{3x+3y}{x^3y^3}$
 - b) $\frac{3x^2}{x^3+y^3}$
 - c) $\frac{3y}{x^3y^3}$
 - d) $\frac{3y^2}{x^3+y^3}$

m) Find $\frac{\partial z}{\partial y}$ where $z = ax^2 + 2by^2 + 2bxy$,

a) $3by$

b) $2ax$

c) $3(ax+by)$

d) $4by+2bx$

SECTION B

Q2 a) Use Taylors theorem to find $\sqrt{25.15}$ **05**

b) Prove that $\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x + 4^x}{4} \right)^{\frac{1}{x}} = (24)^{1/4}$ **05**

Q3 a) If $xu = yz, yv = zx, zw = xy$ then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ **05**

b) If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, prove that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \sin 4u - \sin 2u$ **05**

Q4 a) Solve: $\sin x \frac{dy}{dx} + y \cos x = 2 \sin^2 x \cdot \cos x$ **05**

b) Solve: $\frac{dy}{dx} = \frac{\tan y - 2xy - y}{x^2 - x \tan^2 y + \sec^2 y}$ **05**

Q5 a) Explain $\log(1+\sin x)$ in powers of x . **05**

b) Test the convergence of $\sum_{n=1}^{\infty} \frac{1}{(n+1)^2}$ **05**

Q6 a) If $z = x^y$ then show that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$ **05**

b) Divide 24 into three parts such that the continued product of the first square of the Second & cube of the third is maximum **05**

Q7 a) $3y^2 \frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$ **05**

b) A body of mass m , falling from rest is subjected to the force of gravity & an air resistance proportional to the square of the velocity. If it falls through a distance x & possesses a velocity v at that instant, Prove that $\frac{2kx}{m} = \log\left(\frac{a^2}{a^2 - v^2}\right)$ where $mg = ka^2$. **05**